Calibrated Hallucination Risk Estimation in Large Language Models Using Evidence-Consistent XGBoost

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Abstract

Large language models answer in fluent text, and some answers are unsupported by the evidence a user has. A confidently wrong answer can mislead people in question answering. Current detectors have practical limits. A large judge model is slow and costly, encoder classifiers need model weights and a graphics card, and feature-based detectors rarely report calibration or behavior outside the training domain. This study asks whether a small black-box model can estimate the risk that an answer is hallucinated from the question, the evidence, and the candidate answer alone. A second question is whether that estimate can be trusted as a probability. The proposed method, evidence-consistent XGBoost (EC-XGB), is a gradient-boosted tree model over 35 features. The features cover overlap, entities, numbers, hedging words, semantic similarity, and entailment, and eight of them summarize the support of each atomic claim. Training caps the effect of answer length, keeps the score from falling when the answer contradicts the evidence, and adds natural responses with a source marker. The benchmark is close to saturated, so the gain has to come from reliability under shift, and the natural-answer corpora label that test with grouped splits and three seeds. On HaluEval question answering, EC-XGB holds the in-domain score at F1 = 0.9855. The gain appears under domain shift, although discrimination stays weak. The AUROC rises from 0.475 to 0.582 while the share of combined RAGTruth answers flagged as risky falls from 99.9 to 61.3 percent. The held-out RAGTruth QA task stays near F1 0.30. A display calibrator fitted on RAGTruth cuts the deployed expected calibration error from 0.726 to 0.126. One analysis takes 61.8 ms and costs about USD 0.001 per 1,000 predictions.

1. Introduction

Hallucination detection is an active research area. SelfCheckGPT samples multiple responses and measures their consistency without an external database [1]. Luna uses a small evaluation model that costs less than a large judge [2]. Semantic entropy groups sampled answers by meaning and flags confabulation from disagreement, again with no external knowledge [3]. A hybrid framework combines frozen encoder features with lightweight classifiers and reports strong benchmark scores [4]. Two practical problems remain. Judge models are slow and expensive, and encoder models are hard to interpret. Feature-based detectors are cheaper, but their probabilities are rarely calibrated and their behavior outside the training domain is rarely tested. Three recent surveys group these families and list evaluation and reliability limits among the open questions [5, 6, 7].

Six recent studies are closest to this work. HaluAgent shows that small language models can select tools and detect multiple hallucination types, yet each answer needs multiple tool calls [8]. Valentin et al. report a calibrated multi-scoring detector with a clear cost analysis, and the detector still depends on LLM scoring for its main signal [9]. Sundaragiri et al. combine retrieval grounding, NLI verification, and an XGBoost classifier, and their evaluation reports classification metrics only [10]. Haq et al. score hallucination with token-level SHAP-LIME attributions, without testing whether those attributions stay stable [11]. Deng et al. study cross-domain detection and show that detectors lose accuracy outside their training domain, and their method targets multi-turn dialogue signals [12]. FaithBench reports that modern LLMs still produce hard summarization hallucinations, and it is a summarization benchmark with no risk-calibration study [13].

Four issues remain open in the literature. Calibration is rarely reported for lightweight detectors, explanation quality is rarely measured, cross-domain behavior is rarely tested, and error patterns are rarely analyzed case by case. This study

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addresses the four issues together. It builds a black-box model that reads a question, a reference context, and a candidate answer and estimates $P(\text{hallucinated} \mid q, c, a)$. The model, evidence-consistent XGBoost (EC-XGB), combines 26 evidence-consistency features in seven groups with eight claim-level NLI aggregates and a tuned XGBoost classifier. Platt and isotonic calibration produce the probability, and SHAP explanations show the feature contributions. Group-aware splits and artifact-control baselines cover the benchmark shortcuts, and validation-only calibration answers the reliability question. Zero-shot runs on RAGTruth and FaithBench measure the transfer limits.

HaluEval QA provides the training and in-domain test data, and RAGTruth supplies natural responses for the transfer, multi-source training, and recalibration checks. FaithBench adds a summarization stress test. HaluEval QA is suitable for training because it provides paired correct and hallucinated answers for 10,000 questions. RAGTruth provides natural responses with human span annotations for an external check. The accepted outcomes are a calibrated risk model (EC-XGB) with F1 = 0.9855 on HaluEval QA. Under shift, the flag rate on natural RAGTruth answers falls from 99.9% to 61.3%, while the AUROC rises from 0.475 to 0.582. Target recalibration reduces the RAGTruth QA ECE from 0.8185 to 0.1335. An operating point maximizes recall under a five percent false-positive budget, and a deployed system completes one analysis in 61.8 ms for near $0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

A hallucination is text that is fluent but not supported by the available evidence. Detection quality is usually judged with labeled benchmark splits and the standard metrics of precision, recall, F1, and the area under the receiver operating characteristic curve (AUROC). Standard test beds include HaluEval [14] and TruthfulQA [15], and FActScore [16] measures factual precision in long-form generation. Two evaluation choices matter for deployment. The first is calibration. A risk score is only useful when its numeric value matches the observed frequency of hallucinated answers, and expected calibration error and reliability diagrams expose that gap [17]. The second is interpretability, since a user who sees a high-risk flag needs to know which evidence signal produced it, and feature-attribution methods such as SHAP provide that view for tabular models [18]. The present work evaluates both properties in a black-box setting, where the detector reads the question, an optional reference context, and the candidate answer, and never inspects the generating model.

2.2. Baseline Method

The conventional method in this study is a tuned XGBoost classifier [19] over the 26 hand-crafted evidence-consistency features. XGBoost is a gradient-boosting ensemble, so the model adds decision trees one at a time and each new tree fits the error left by the trees already in the ensemble [19]. Each tree splits the feature space into leaves, and each leaf stores a weight that adds to the running score. The final score passes through a logistic link, so the output is a probability of the hallucinated class.

Let x_i be the feature vector of row i and $y_i \in \{0, 1\}$ its label. The training objective combines a differentiable loss with a regularizer over the trees,

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \left(\gamma T_k + \tfrac{1}{2} \lambda \|w_k\|^2 \right), \tag{1}$$

where f_k is the k-th tree with T_k leaves of weight w_k, a leaf-count penalty γ, and an L2 weight penalty λ. The loss is the standard logistic loss and the probability comes from the sigmoid of the summed leaf scores, $\hat{y}_i = \sigma(z_i) = 1/(1+e^{-z_i})$.

Tree depth, learning rate, tree count, subsample rate, and column subsample rate come from a randomized search over 30 configurations, scored with grouped 5-fold cross-validation on the training split. The selected values are listed with the experimental setup in Section 4. This baseline is the reference for every later modification, and its training and inference procedure is summarized in Algorithm 1. The procedure extracts the features once, tunes the hyperparameters with a grouped search, and repeats training, calibration, and evaluation for each seed so that the reported metrics are seed-averaged.

Algorithm 1 Baseline feature-based XGBoost (training and inference)

Require: labeled rows (q_i, c_i, a_i, y_i) and seeds $S = \{42, 123, 456\}$
Ensure: trained model, calibrator, and decision threshold
1: extract the 26 evidence-consistency features for every row
2: tune XGBoost with a randomized search and grouped 5-fold cross-validation
3: **for** each seed $s \in S$ **do**
4: train the classifier on the training split
5: fit Platt calibration and select the decision threshold on the validation split
6: evaluate on the fixed test split
7: **end for**
8: **return** the mean metrics over the seeds

2.3. Limitations and Research Gap

The baseline works well on its own benchmark but has clear limits. On HaluEval the benchmark is close to saturated, so a better feature set buys almost no accuracy, and an in-domain win is a weak target. The model also leans on length and lexical overlap, and the error analysis in Section 5 shows that answer length alone is close to a perfect cue in this corpus. Under domain shift the raw probabilities turn unreliable, since the model flags almost every natural answer and its confidence no longer matches the observed error rate.

These limits connect to open problems in the literature. Token-level attributions explain raw text rather than semantically meaningful evidence signals, and they are visualized without any reliability measurement [11]. Calibration is rarely reported for lightweight detectors, and target-domain behavior is rarely tested, while the closest multi-signal framework reports classification metrics only [10]. The benchmark itself imposes limits as well, since around 85% of HaluEval is generated with a sampling-then-filtering procedure [20], and such data blurs the line between factually wrong and hallucinated content. Cross-domain evaluation stays open, and SpikeScore shows that detectors trained in one domain lose accuracy in another [12]. This study fills these gaps by fusing claim-level evidence signals and multi-source training into an evidence-consistent XGBoost and by testing calibrated, explainable risk prediction under domain shift with a reproducible artifact.

3. Proposed Method

3.1. Working Principle and Task

Figure 1 shows the full proposed pipeline at a glance. The system takes a question, an optional reference context, and a candidate answer produced by a black-box LLM, and it returns a calibrated risk probability together with per-feature explanations. Each stage is described in the following subsections.

Given a question q, an optional reference context c (evidence), and a candidate answer a produced by a black-box LLM, the task is to predict the probability $P(\text{hallucinated} \mid q, c, a)$. The proposed model predicts risk, not truth. It flags answers that are unsupported by, or contradictory to, the available evidence.

3.2. Features

Table 1 lists the 26 base features in seven groups, and Table 2 lists the eight claim-level aggregates that the proposed model adds. Natural language inference (NLI) decides whether one sentence follows from another, and the NLI signals come from a DeBERTa-v3 cross-encoder [21] trained on SNLI [22] and MultiNLI [23], evaluated in both directions, context-to-answer and answer-to-context, which yields six probability features. Semantic similarity uses MiniLM sentence embeddings [24, 25], entity features use spaCy NER [26], and the numeric and hedging groups are collected with regular expressions and a small hedging lexicon.

3.3. Evidence-Consistent XGBoost

EC-XGB keeps the standard XGBoost learner and changes three things, added cumulatively so that each change can be measured on its own. The first change replaces the raw length counts with log-scaled values and adds monotone constraints, so risk cannot fall when contradiction or novelty rises and cannot climb when entailment, overlap, or semantic similarity rises. A log-scaled length feature replaces the raw count, $\ell' = \log(1 + \ell)$, which compresses the long tail of the length distribution. A monotone constraint fixes the sign of the model response for a chosen feature, so an evidence-positive feature x_j keeps $\partial\hat{y}/\partial x_j \leq 0$, and the sign flips for an evidence-negative feature such as contradiction. In plain terms, more contradiction can never lower the risk score, and more entailment can never raise it. The second change appends the eight claim-level aggregates from Table 2, which lets the classifier see the same evidence signals

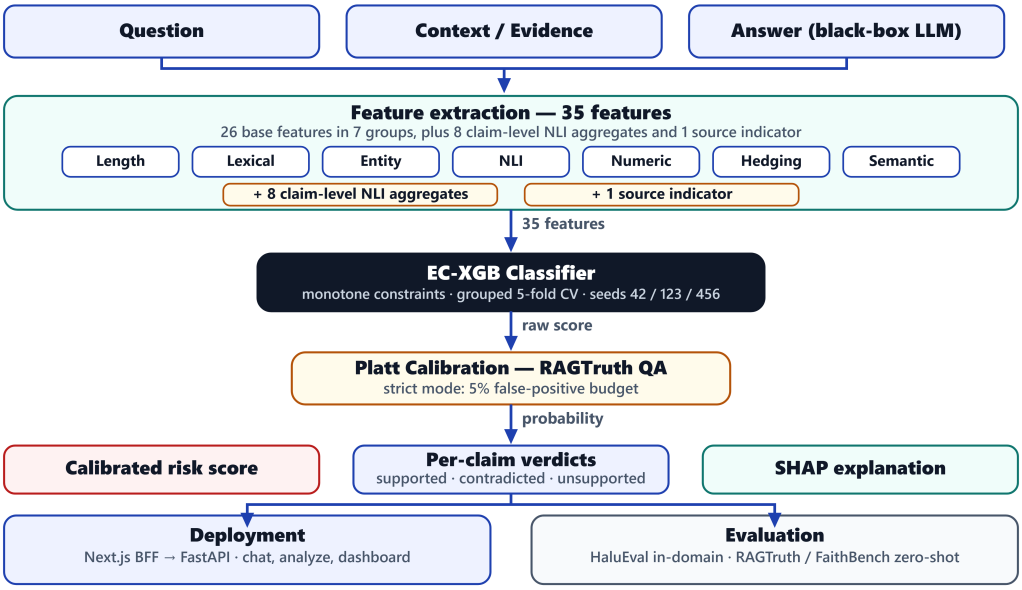


Figure 1: End-to-end architecture of the proposed model. The inputs feed 35 features into the EC-XGB classifier, Platt calibration produces the deployed score, and the outputs are the risk score, the per-claim verdicts, and the SHAP explanations. The system is deployed as a Next.js and FastAPI application and evaluated in-domain on HaluEval and under shift on RAGTruth and FaithBench.

Table 1
Base feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard ×2	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral × 2 directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

Table 2
Claim-level NLI aggregates (8 features). The answer is split into at most six atomic clauses, with conjunctions such as "and" or "but" broken into separate clauses, each clause is scored against its four best-matching context sentences, and the pair probabilities are aggregated.

Signal	Features	Source
Contradiction	max and mean contradiction, contradicted-claim ratio	NLI cross-encoder
Support	supported-claim ratio, unsupported-claim ratio, min and mean entailment	NLI cross-encoder
Decomposition	number of claims	sentence splitter

that the deployed verification layer reports. For an answer split into K claims with per-claim contradiction c_k and entailment e_k, the aggregate vector and the full model input are

$$\psi(x) = \left[\max_k c_k,\ \bar{c},\ r_{\text{con}},\ r_{\text{sup}},\ r_{\text{unsup}},\ \min_k e_k,\ \bar{e},\ K\right], \tag{2}$$

$$\phi'(x) = \left[\ell',\ \tilde{\phi}(x),\ \psi(x),\ s\right], \tag{3}$$

where r_{con}, r_{sup}, and r_{unsup} are the contradicted, supported, and unsupported claim ratios, and $\tilde{\phi}(x)$ holds the remaining base features after the length change, which gives 35 inputs in total. The third change adds RAGTruth summarization and data-to-text training rows with a binary source indicator $s \in \{0, 1\}$, while every RAGTruth QA row stays out of training, which keeps the QA transfer result zero-shot for the multitask model. The deployed checkpoint combines all three changes and is called EC-XGB. A strict decision mode selects the validation threshold whose false-positive rate stays at or below five percent, and the calibration protocol below fits its display score. Algorithm 2 lists the full procedure: it trains one model per seed, fits the display calibrator on the RAGTruth QA calibration split, selects the strict threshold on the HaluEval validation split, and returns the deployed artifacts.

Algorithm 2 Evidence-consistent XGBoost (EC-XGB) training and deployment

Require: labeled HaluEval QA rows, RAGTruth non-QA rows, seeds S
Ensure: deployed EC-XGB model, display calibrator, and strict threshold
1: extract the 26 base features and the eight claim-level aggregates
2: log-scale the length features and set the monotone constraints
3: **for** each seed $s \in S$ **do**
4: build the 35-feature input with the source indicator
5: train XGBoost on HaluEval plus RAGTruth non-QA rows, and keep every RAGTruth QA row out of training
6: fit the display calibrator on the RAGTruth QA calibration split
7: select the strict threshold on the HaluEval validation split
8: **end for**
9: **return** the deployed model, calibrator, and threshold

The contribution is the combination. Each piece is standard on its own, and together they hold the in-domain score, cut the false-alarm rate under shift, and keep the output explainable at the feature level. The deployed system adds claim-level verification with evidence retrieval and a display calibrator fitted on the target domain. Relative to the closest feature-based detector [10], the difference is the claim-level evidence features and the calibration protocol. Relative to the LLM-scored detector [9] and the tool-based detector [8], the difference is that inference needs no model calls.

3.4. Calibration and Explainability

Calibrators are fit on the validation split only. Platt scaling fits a sigmoid $p(x) = \sigma(Af(x) + B)$ to the raw score $f(x)$, where A and B are fit by maximum likelihood [27]. Isotonic regression fits a non-decreasing step function g so that $p(x) = g(f(x))$ [28]. Both are compared against the raw model on the test split. The comparison uses expected calibration error over $M = 10$ bins,

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|, \tag{4}$$

where $\text{acc}(B_m)$ and $\text{conf}(B_m)$ are the accuracy and mean confidence in bin B_m, together with the Brier score $\text{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$ and reliability diagrams [17]. A second step targets a new domain. The calibrator can be re-fit on a small labeled sample of the target corpus, and the effect is measured on a disjoint held-out part of the target corpus. The deployed EC-XGB display score follows the same rule: its calibrator is fitted on the designated RAGTruth QA calibration split, and the source indicator is set to one for natural responses, the same convention used in the shift evaluation.

Global interpretability comes from SHAP [18]. A TreeExplainer over the raw XGBoost model produces mean absolute SHAP importance and beeswarm summaries, and local explanations show the feature contributions behind a single verdict. Explanation reliability is measured directly in this work, since feature-attribution methods can be unstable [29]. Each test answer is perturbed in six ways, by replacing entities, numbers, or dates, removing supporting context, inserting irrelevant context, and shuffling clauses, and the calibrated score should stay stable under neutral perturbations and react to semantic ones. Stability is quantified with Spearman and Kendall rank correlation between the original and perturbed scores, the fraction of top-1 flips, and the top-k Jaccard overlap of the SHAP feature rankings.

3.5. System and Reproducibility

The methods above are deployed as a two-tier web application. The Next.js frontend streams an LLM conversation on the chat page using the assistant-ui library, and the FastAPI backend holds the model, the calibrator, the SHAP explainer, and the feature models, all loaded once at startup. Every machine-learning request from the frontend goes through a backend-for-frontend proxy in the Next.js layer to the FastAPI endpoints, which avoids cross-origin issues and keeps the API key on the server side. All quantitative results in Section 5 measure the EC-XGB classifier and its calibrator alone. The claim-verification layer described below is a separate deployment component. It changes the displayed verdict and is demonstrated through the interface rather than measured in the reported metrics.

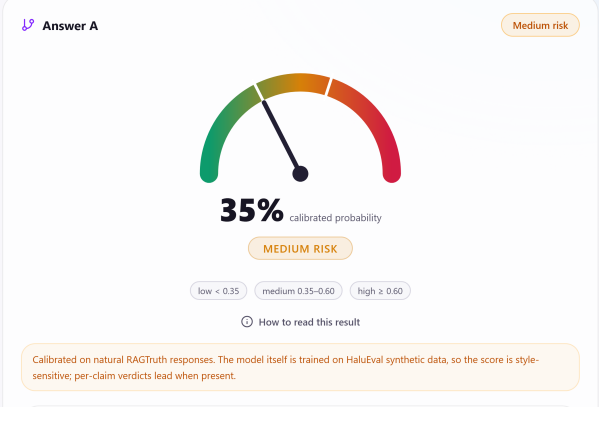
The chat page streams through the Vercel AI SDK route. A tool call from the conversation model triggers the analysis endpoint, and the result renders as a risk card with the calibrated score, the per-claim verdicts, the SHAP bars, and an evidence panel.

Beyond the score, the system verifies the answer claim by claim. Each claim is decomposed from the answer text and checked with sentence-level NLI against one of three evidence sources, the provided context, an indexed document collection, or live web search. Verdicts are decided support-first, so any entailing sentence supports a claim and a contradiction decides only when no sentence supports it, and a relevance gate drops retrieved sentences that do not cover the claim vocabulary. The verdicts supported, contradicted, and unsupported are rendered per claim, with the driving evidence sentence quoted for contradicted claims. Web search uses the Brave Search API with its LLM-context endpoint, and the document index combines BM25, FAISS dense retrieval, and reciprocal rank fusion with a cross-encoder reranker. An uncertain verification score routes the claim to an LLM judge that returns a verdict with a short reasoning note.

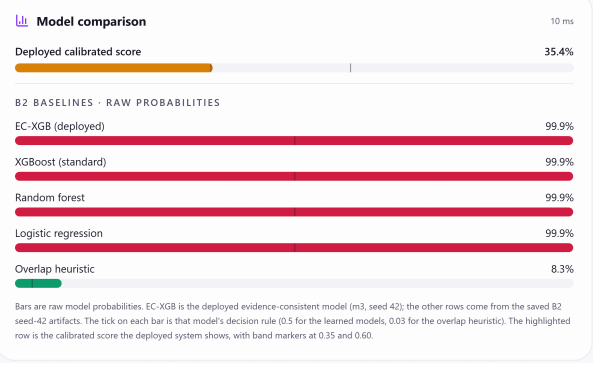
The verdict headline leads the card. Any contradicted claim sets the risk to high, an unsupported claim sets it to medium, and fully supported claims set it to low, while the standard XGBoost score appears beneath it as the legacy model score. This ordering follows the style-sensitivity result in Section 5, where a full-sentence grounded answer can saturate the calibrated score even though every claim is supported. The displayed percentage comes from the EC-XGB display calibrator on the RAGTruth QA split and is adjusted by the claim verdicts.

The analyze page shows the risk gauge, the calibrated score, the SHAP bars, and a comparison against the standard model.

Figure 2 shows the analyze page for the hallucinated-number case, where the context states 5 days per decade and the answer states 10. The raw EC-XGB probability is 0.999, so the raw score alone is not an actionable risk level, while the deployed display calibrator reports 35.4% and marks the answer as medium risk. The comparison panel shows the same pattern across the baselines, with the learned models all near 99.9% and the overlap heuristic at 8.3%, which is why the interface leads with the calibrated score and keeps the raw score as a secondary signal.



(a) Risk card: the calibrated display score is 35.4% (medium risk).



(b) Model comparison: raw probabilities of the deployed model and the baselines.

Figure 2: Analyze page for the hallucinated-number case, where the context states 5 days per decade and the answer states 10. The risk card shows the calibrated display score, and the comparison panel shows the raw probability of each model. The raw EC-XGB probability is 0.999, while the deployed display calibrator reports 35.4%, which makes the risk level actionable.

The source repository contains the complete pipeline. `src/data` holds the download and split scripts, `src/features` the seven feature groups, `src/models` the training protocol together with the cross-domain, calibration-shift, and explanation-reliability runs, `src/explain` the SHAP analysis, and `src/api` the FastAPI application. `requirements.txt` pins every dependency with exact versions. The train, validation, and test split indices are saved to `artifacts/split_indices.json` and `.npy`, all seeds are centralized in the run configuration, and the model, calibrator, scaler, and SHAP explainer artifacts are stored under `artifacts/models/`.

The full experiment protocol is reproducible end to end. A self-contained Colab notebook (`colab/HaluRISC_Training_Version_B.ipynb`) embeds the runtime source, verifies file hashes before execution, runs the pipeline on a GPU, and persists the artifacts to Google Drive. The `run_all_experiments.py` script reproduces the B-run locally from a frozen configuration (`configs/version_b.yaml`), `make_manifest.py` freezes the artifact hashes into `manifest.frozen.json`, and `verify_artifacts.py` checks that every reported number matches the frozen artifacts. The manuscript itself is built from the same files that the dashboard reads, so the tables in Section 5 trace back to the artifact files, including the length-binned error table and the reviewed audit sheet.

4. Experiments

4.1. Experimental Setup and Conditions

All experiments run on a Windows 11 laptop with an NVIDIA RTX 3060 6 GB GPU (CUDA 12.8, torch 2.11.0+cu128) and 32 GB of RAM, and the code runs on Python 3.12. The model uses scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. The XGBoost parameters are depth 4, learning rate 0.01, 500 trees, subsample 0.9, and column subsample 0.7, selected by the randomized search described above. Every run is fixed by the saved split indices and the centralized seeds, so the same command reproduces the same tables.

4.2. Dataset

The QA subset of HaluEval [14] is the training source. It contains 10,000 questions, each with a knowledge passage, a correct answer, and a hallucinated answer, which gives 20,000 labeled rows. The rows are split 70/15/15 at the question-group level, producing 14,000 training, 3,000 validation, and 3,000 test rows. This grouped, leakage-free design matters because both answer variants of a question share the same context, and a random split could otherwise let the model memorize the pairing. The split indices are saved for exact reproducibility.

Transfer is measured on RAGTruth [30], a corpus of naturally generated responses with human word-level span annotations, and on FaithBench [13], a summarization hallucination benchmark built from modern LLM outputs. The RAGTruth QA task contributes 5,934 rows in 989 groups, of which 5,034 rows (839 groups) form the recalibration set and 900 rows (150 groups) the held-out test, with no overlapping groups between the two. Table 3 summarizes the corpora.

Table 3
Datasets used for training and evaluation.

Corpus	Rows	Groups	Use
HaluEval QA	20,000	10,000	train / validation / test
RAGTruth QA	5,934	989	held-out + recalibration
RAGTruth full	17,790	2,965	transfer (seen tasks)
FaithBench	750	750	zero-shot

Text is whitespace-normalized. Lexical features operate on lowercased token sets, while NER and NLI keep the original casing. Missing context is handled explicitly, with lexical overlap set to zero and NLI probabilities to one third. All three corpora are used under their original licenses: HaluEval and RAGTruth are MIT-licensed, and FaithBench is distributed under CC BY-NC-SA 4.0. Raw files are not redistributed. The repository ships the download scripts, source revisions, and SHA-256 hashes instead, so every split can be regenerated from the original repositories.

4.3. Evaluation Protocol

Eight models are compared. The overlap heuristic sets risk to one minus the lexical overlap between the answer and the context, and its threshold is tuned on validation. Logistic regression on standardized features and a random forest with 300 trees are the two standard classifiers. The standard classifier is XGBoost [19], trained on the 26 base features,

and it is the reference for the modified model in Section 3. Three controls isolate the value of the feature design, an NLI-only logistic regression that sees only the six NLI probabilities, an answer-only TF-IDF linear model, and a TF-IDF linear model over the full input. Internal-state detectors read hidden activations and need model access, so they sit outside the black-box setting [31]. A majority baseline that predicts the negative class for every row confirms the balance of the labels. XGBoost hyperparameters come from a randomized search of 30 iterations over depth, learning rate, tree count, subsample, and column subsampling, scored with 5-fold stratified cross-validation on the training split. The selected configuration gives a cross-validated AUROC of 0.9970, and the values are listed with the experimental setup in Section 4. Every experiment is repeated with seeds 42, 123, and 456, and results are reported as the mean over seeds.

Evaluation uses precision, recall, F1, AUROC, the area under the precision-recall curve (PR-AUC), and the Matthews correlation coefficient (MCC), and calibration with expected calibration error over 10 bins, Brier score, and negative log-likelihood. Metric tables report four decimals, and latency is reported in milliseconds. Hyperparameters are chosen by a randomized search of 30 iterations with grouped 5-fold cross-validation on the training split. Every experiment is repeated with seeds 42, 123, and 456 and reported as the mean over seeds. Significance is assessed with McNemar's test on the paired seed-42 test predictions, bootstrap 95% confidence intervals from 1,000 resamples over question or source groups, and a Wilcoxon signed-rank test across seeds. The group structure is respected everywhere, so the two rows of one question never appear in both the training and the evaluation folds.

4.4. Additional Protocols

Calibrators are fit on the validation split only, and the comparison of raw, Platt-scaled, and isotonic probabilities happens on the test split. Calibration is known to degrade under distribution shift [32], so the target-domain protocol follows the same principle. The recalibration set of 5,034 RAGTruth QA rows is used only to fit the target calibrator, and all reported numbers come from the disjoint 900-row test set.

Global explanations come from a SHAP TreeExplainer [18] over the raw XGBoost model. Three measurements check whether the explanations are trustworthy. First, the mean absolute SHAP ranking is compared against permutation importance with a Kendall tau correlation. Second, the top-5 feature set is re-estimated from 1,000 bootstrap resamples and compared with the Jaccard index. Third, two perturbation studies are run on the test answers. Content edits replace entities, numbers, or dates, remove the supporting context, or insert an irrelevant sentence, and each edit should move the calibrated score if it changes the evidence. Structure edits shuffle the clause order, and these should leave the score nearly unchanged. For each operator the mean absolute score change, the fraction of large score changes, the top-1 SHAP flip rate, and the Spearman correlation between the original and perturbed scores are recorded. A neutralization study sets the top-k SHAP features of each sample to its median and tracks the mean score change for $k = 1, 3, 5, 10$.

Latency is measured on 200 test samples on the reference laptop with an RTX 3060 6 GB GPU, where NLI and embeddings run on the GPU and the remaining features on the CPU. The LLM-as-judge comparison uses GPT 5.6 Luna, a standard but biased comparator [33], on the same 200 samples with a fixed prompt, and the cost is computed from the recorded token counts.

5. Results

5.1. Model comparison

Table 4 reports the test-set performance of every baseline, averaged over the three seeds. XGBoost reaches an F1 of 0.9846 with an AUROC of 0.9979, ahead of the random forest by 0.0004 in F1 and the logistic regression by 0.0186. The overlap heuristic, which needs no training beyond its threshold, still reaches an F1 of 0.9352, and a majority baseline that predicts the negative class for everything scores zero, which confirms that the labels are balanced. The NLI-only baseline lags at 0.6714, which shows that the lexical, length, and semantic signals carry information the NLI cross-encoder alone does not. The TF-IDF variants sit between these extremes, with an answer-only linear model at 0.9224 and a full-input model at 0.5999. The margin between XGBoost and the random forest is small but significant in a McNemar test on the paired seed-42 test predictions ($p = 0.044$), while the comparison against logistic regression ($p = 0.43$) and the heuristic ($p = 0.35$) is not significant. The Wilcoxon test across the three seeds finds no difference, which is expected given that the test compares only three points. Bootstrap confidence intervals over 1,000 question-group resamples for XGBoost are F1 [0.9809, 0.9896] and AUROC [0.9964, 0.9989]. The standard XGBoost row in

Table 4 and the m0 row in Table 5 come from the baseline and modified pipelines. The two runs differ only within seed-level variation (F1 0.9846 against 0.9844).

The split design is verified against a known confound. A row-level split, where the two rows of one question can fall into different partitions, inflates F1 to 0.9886, and the grouped split corrects this to 0.9846, a drop of 0.004 in F1 and 0.0001 in AUROC. The reported numbers all use the grouped split.

Table 4
Test-set performance on HaluEval QA, mean over seeds 42/123/456 (n = 3,000). The ECE and Brier columns measure the reliability of each model's risk score. The majority baseline predicts the negative class for every row. Probability-based metrics are undefined for a constant score.

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1−overlap)	0.9473	0.9233	0.9352	0.9132	0.8196	0.8723	0.3541	0.2672
Logistic regression	0.9768	0.9553	0.9660	0.9932	0.9901	0.9329	0.0099	0.0253
Random forest	0.9897	0.9789	0.9842	0.9980	0.9983	0.9687	0.0143	0.0127
NLI-only	0.6479	0.6967	0.6714	0.7177	0.7531	0.3189	0.0678	0.2182
TF-IDF (answer-only)	0.9550	0.8920	0.9224	0.9696	0.9665	0.8519	0.0831	0.0662
TF-IDF (full input)	0.6397	0.5647	0.5999	0.6754	0.6785	0.2484	0.0679	0.2328
XGBoost (standard)	0.9932	0.9762	0.9846	0.9979	0.9983	0.9697	0.0045	0.0117
Majority (negative)	0.0000	0.0000	0.0000	—	—	0.0000	—	—

Single-answer behavior shows where aggregate metrics hide disagreements: for a grounded span and a hallucinated sentence that share one question and one context, the overlap heuristic and the NLI-only model fail to separate the pair, while logistic regression, random forest, and XGBoost separate it cleanly. A grounded one-word answer is flagged by every raw model, and the deployed calibrator maps those raw probabilities to 0.0000.

5.2. Evidence-consistent XGBoost

Table 5 reports the component ablation for EC-XGB on the HaluEval test set, averaged over the three seeds. The standard XGBoost reference (m0) reaches an F1 of 0.9844, the length debias and monotone constraints (m1) reach 0.9852, the claim-level aggregates (m2) reach 0.9857, and the full EC-XGB (m3) reaches 0.9855 with an MCC of 0.9715. The differences are small and not significant in a McNemar test on the paired predictions (m0 against m3, $p = 0.45$), and a Wilcoxon test across seeds finds no difference either ($p = 0.25$). The benchmark is saturated, so this ablation is a preservation check rather than a win. Every component holds the line. On the seed-42 test split, the false-positive count falls from 12 to 8. Removing the source indicator from the same recipe changes the in-domain F1 by less than 0.001 (0.9849 against 0.9855) and the shifted flag rate by less than one point, so the multi-source gain does not rest on source identity. At inference, the indicator is set to one for natural responses and zero for benchmark-style rows.

Table 5
EC-XGB component ablation on the HaluEval test set, mean over seeds 42/123/456 (n = 3,000). m0 is the standard XGBoost reference, m1 adds log-scaled length features and monotone constraints, m2 adds the eight claim-level aggregates (34 features), and m3 adds the binary source indicator (35 features) and is the deployed EC-XGB.

Variant	Features	F1	AUROC	MCC	ECE
m0 standard	26	0.9844	0.9980	0.9693	0.0066
m1 debias	26	0.9852	0.9976	0.9708	0.0051
m2 + claims	34	0.9857	0.9984	0.9717	0.0078
m3 EC-XGB	35	0.9855	0.9984	0.9715	0.0074

Table 6 shows where the modification matters. Every model is applied without retraining to the RAGTruth and FaithBench corpora. The standard model flags almost every natural answer as risky, since 99.9% of RAGTruth rows and 99.5% of FaithBench rows cross the 0.5 threshold, which makes the decision rule useless outside the benchmark even though recall stays high. EC-XGB cuts the flag rate (the share of answers above the 0.5 threshold) to 61.3% on the RAGTruth test tasks and to 7.3% on FaithBench, raises the AUROC from 0.475 to 0.582 and from 0.509 to 0.594, and lowers the ECE on RAGTruth from 0.635 to 0.278. The held-out RAGTruth QA task is the strictest test, since its rows never enter training. There the F1 stays flat (0.3022 against 0.3081) and the ECE improves from 0.815 to 0.743, which shows that multi-source training does not transfer a new task for free. On FaithBench the conservative regime lowers the F1 from 0.813 to 0.120. The standard model earns that F1 by flagging nearly every row, so the AUROC and the flag rate are the fair comparison at this operating point, and the F1 drop is reported as a limitation rather than hidden.

Table 6
Cross-domain behavior, mean over seeds. Standard is the B2 XGBoost trained on HaluEval only. Its external results are zero-shot. EC-XGB adds the RAGTruth non-QA training rows, so its summarization and data-to-text results measure adaptation, while its QA results are held out. RAGTruth all uses the 2,700-row test split, RAGTruth QA uses the 900-row held-out split, and FaithBench uses all 750 rows. Flagged is the share of rows above the 0.5 threshold.

Corpus	Model	F1	AUROC	Flagged	ECE
RAGTruth all	Standard	0.5181	0.4753	99.9%	0.6346
RAGTruth all	EC-XGB	0.5608	0.5815	61.3%	0.2782
RAGTruth QA	Standard	0.3022	0.5465	99.9%	0.8149
RAGTruth QA	EC-XGB	0.3081	0.5491	97.4%	0.7433
FaithBench	Standard	0.8130	0.5089	99.5%	0.2917
FaithBench	EC-XGB	0.1201	0.5941	7.3%	0.4500

The operating-point mode selects the threshold on the HaluEval validation split that maximizes recall under a five percent false-positive budget. The test false-positive rate stays near 5% while recall stays above 99%, a recall-first option at about one point of F1. For the deployed display score, a Platt calibrator fitted on the 5,034-row RAGTruth QA calibration split reduces the ECE on the held-out 900-row QA test from 0.726 to 0.126. This calibrator is fitted on EC-XGB scores and is the one used by the deployed interface. Table 8 reports the same protocol for the standard model, whose raw score is more saturated (0.8185) and whose target Platt calibrator reaches 0.1335. The gap to 0.126 therefore comes from the upstream model, not from the protocol.

5.3. Calibration

Table 7 compares raw, Platt-scaled, and isotonic probabilities on the HaluEval test set. The raw model is already well calibrated with an ECE of 0.0045, and neither recalibration improves it, since Platt raises the ECE to 0.0091 and isotonic to 0.0070. This is an honest negative result: the logistic objective already produces reliable probabilities on this benchmark, and the reliability diagrams in Figure 3 confirm that the raw curve tracks the diagonal.

Table 7
Calibration on the HaluEval test set (mean over seeds).

Method	F1	ECE	Brier	NLL
Raw XGBoost	0.9846	0.0045	0.0117	0.0449
Platt (sigmoid)	0.9848	0.0091	0.0129	0.0579
Isotonic	0.9844	0.0070	0.0124	0.0727

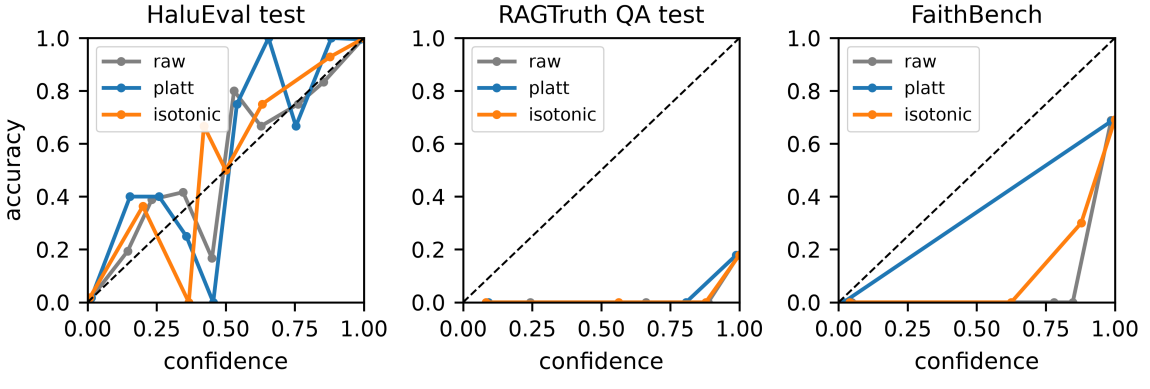


Figure 3: Reliability diagrams of the raw, Platt-calibrated, and isotonic models on the HaluEval test set (left), RAGTruth QA (middle), and FaithBench (right).

Table 8 applies the standard model to the 900-row RAGTruth QA test set and compares raw scores, the source calibrator fitted on the HaluEval validation split, and calibrators re-fitted on 5,034 RAGTruth rows. The test set is 17.8% positive, and the raw and source-calibrated probabilities are badly off, with an ECE above 0.8. Target recalibration cuts the ECE to 0.1335 (Platt) and 0.1316 (isotonic) and the Brier score from 0.801 to 0.164. The price is visible in the same numbers: every calibrated score falls below 0.5 (the maximum is 0.313), so the fixed threshold predicts no positives.

A threshold selected on the calibration split at a five percent false-positive budget (0.313) recovers a test recall of 5.0% at a 5.8% false-positive rate (precision 15.7%, F1 0.076), so target recalibration fixes the probability scale, not the ranking. The labels are cheap, however: 5% of the calibration split (252 rows) already gives a test ECE of 0.133, the same as the full 5,034 rows. Figure 4 shows the raw and target-calibrated score distributions behind these numbers.

Table 8
Target-domain recalibration of the standard XGBoost model on RAGTruth QA (calibration n = 5,034, test n = 900).

Calibrator	ECE	Brier	NLL	Pred. pos.
Raw scores	0.8185	0.8164	5.20	0.999
Source Platt (HaluEval)	0.8089	0.8011	3.62	0.999
Target Platt (RAGTruth)	0.1335	0.1639	0.51	0.000
Target isotonic	0.1316	0.1663	0.53	0.000

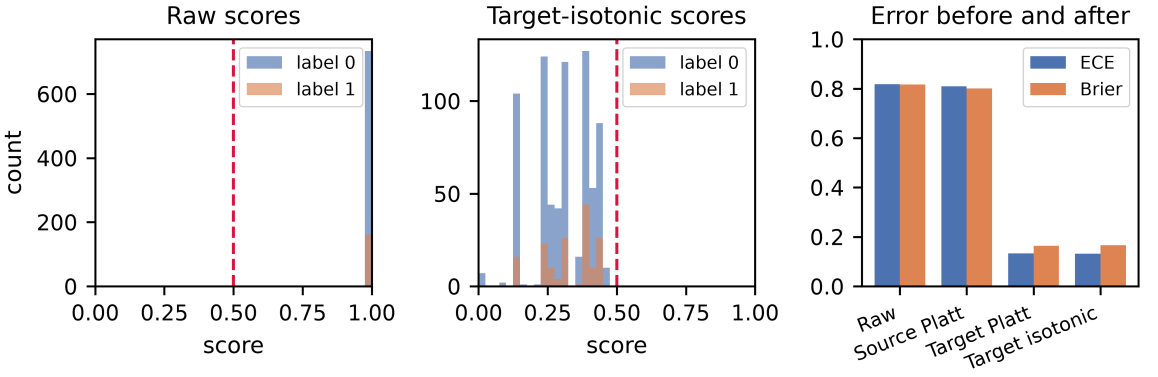


Figure 4: Target-domain recalibration of the standard XGBoost model on RAGTruth QA. Left and middle: raw and target-calibrated score distributions on the 900-row test set, with the 0.5 threshold marked. Right: ECE and Brier before and after recalibration.

5.4. Cross-domain transfer

The transfer evaluation is the central negative result of the paper. The standard model is applied to every external corpus without training or adaptation, so its results there are a clean zero-shot measurement. EC-XGB has seen the RAGTruth summarization and data-to-text training rows, so only its held-out QA and FaithBench results are zero-shot. In the dedicated cross-domain run, the standard model reaches an F1 of 0.302, an AUROC of 0.540, and an ECE of 0.819, and recall is 1.0 on every corpus: the model flags almost everything, the precision lands on the label prior, and the AUROC stays near 0.5. Synthetic HaluEval hallucinations differ from natural responses. FaithBench sits between the two extremes at an F1 of 0.8130 and an AUROC of 0.5314, since its summaries are shorter and closer to the HaluEval format, and the full 17,790-row RAGTruth corpus lands at an F1 of 0.603, an AUROC of 0.497, and an ECE of 0.560. On the full task populations, the breakdown repeats the pattern, with F1 of 0.4515 on QA, 0.4603 on summarization, and 0.8140 on data-to-text, and the F1 rises from 0.4940 on very short contexts to 0.7091 on contexts with 128 to 511 characters. Table 6 reports the EC-XGB comparison on the 2,700-row test split, whose lower positive rate (35% against 43%) explains the lower level. The direction matches. Group-aware bootstrap intervals (1,000 resamples) keep the held-out QA AUROC near chance for both models ([0.514, 0.607] for the standard model and [0.502, 0.603] for EC-XGB). Figure 5 plots the score distributions behind these numbers.

5.5. Ablation

Table 9 removes each feature group and retrains from scratch with the same protocol. Lexical features dominate, with the F1 dropping by 0.0247 when they are removed, while length and NLI contribute smaller amounts and the entity, numeric, and hedging groups stay neutral within 0.0002. Removing the semantic group slightly improves the F1, a negative result reported rather than hidden, and the ranking agrees with the SHAP analysis next.

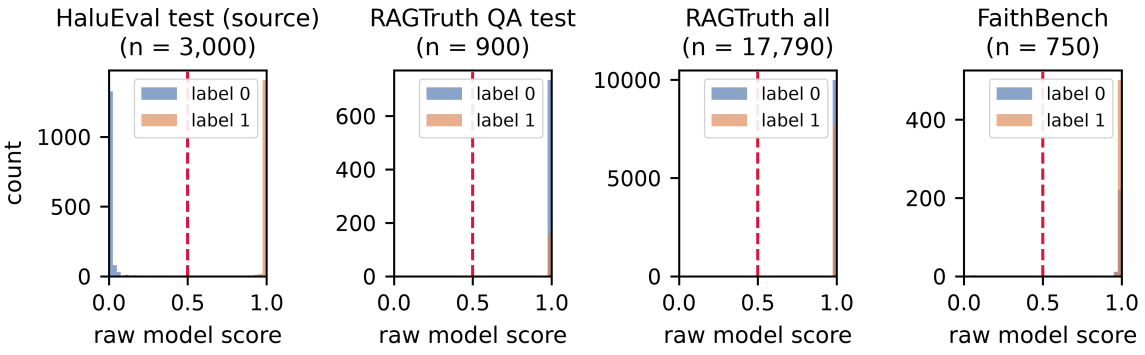


Figure 5: Raw model score distributions on the HaluEval test set and the external corpora, with the 0.5 threshold marked. The source panel separates the classes, while every external panel piles both labels at 1.0.

Table 9
Feature-group ablation, mean over seeds (full model F1 = 0.9846, AUROC = 0.9979).

Removed group	F1	ΔF1	AUROC
length	0.9801	−0.0045	0.9963
lexical	0.9600	−0.0247	0.9923
entity	0.9847	+0.0001	0.9982
nli	0.9823	−0.0023	0.9973
numeric	0.9844	−0.0002	0.9982
hedging	0.9848	+0.0001	0.9981
semantic	0.9860	+0.0014	0.9982

5.6. Explanation reliability

Figure 6 shows the mean absolute SHAP values for the top-12 features. The context overlap dominates, followed by the answer length features, the context-to-answer contradiction probability, and the question-to-answer cosine. The explanation set agrees with the ablation in spirit, since lexical and length features carry the largest weights. Three checks assess whether the explanations are stable. The SHAP ranking agrees with permutation importance at a Kendall tau of 0.66. The top-5 feature set is identical across 1,000 bootstrap resamples, with a Jaccard of 1.0. Neutralizing the top-k features drops the mean score monotonically from -0.079 at $k = 1$ to -0.412 at $k = 10$, which confirms that the top features are actually driving the predictions rather than decorating them.

Table 10 reports the perturbation study. Content edits move the score in proportion to the semantic weight of the edit. Replacing an entity changes the score by 0.360 on average, a number by 0.345, a date by 0.279, and removing the supporting context by 0.272. The irrelevant insert and clause shuffle leave the score nearly untouched, at 0.006 and 0.001. The top-1 SHAP feature never flips under any operator, and the Spearman correlation stays between 0.84 and 0.95. The sample sizes are small for dates and clause shuffles, 13 and 14 respectively, because not every answer contains an edit-able element, and the table reports n openly.

Table 10
Perturbation stability of the calibrated score and SHAP ranking.

Operator	n	Δscore	Large rate	Top-1 flip	Spearman
Entity replacement	78	0.360	0.372	0.000	0.840
Number replacement	22	0.345	0.364	0.000	0.852
Date replacement	13	0.279	0.308	0.000	0.870
Support removal	66	0.272	0.288	0.000	0.874
Irrelevant insert	120	0.006	0.008	0.000	0.948
Clause shuffle	14	0.001	0.000	0.000	0.921

A structured expert audit tested these explanations case by case. The B5 run exported 26 error-enriched cases (9 false positives, 10 false negatives, and 7 inside the 0.35 to 0.65 band), and two review passes were performed with an AI assistant, which the released sheet records in a review-mode field. Both passes judged the explanation against

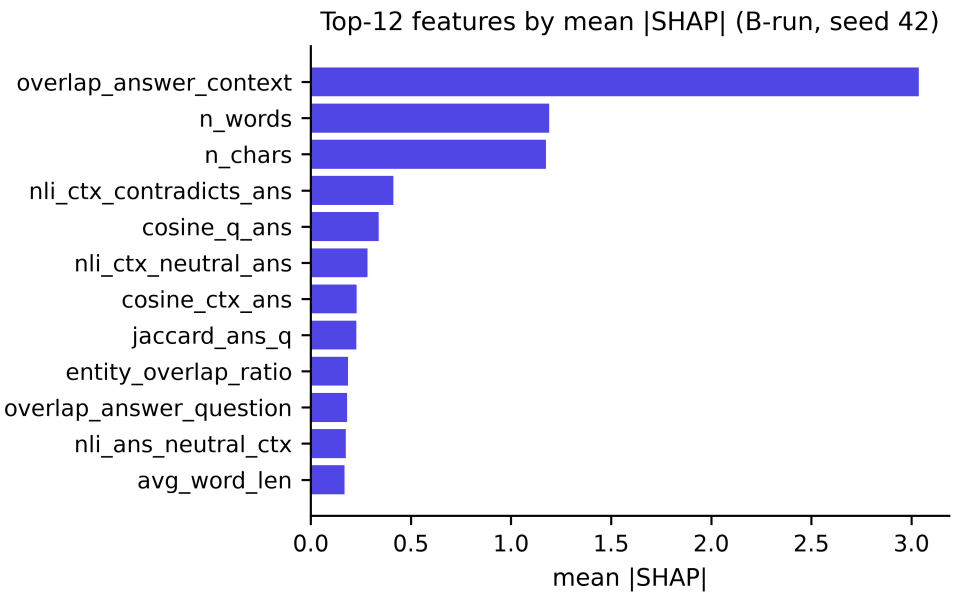


Figure 6: Mean absolute SHAP values for the top-12 features (B-run, seed 42).

the answer-context evidence rather than the dataset label, they agreed on 23 of 26 cases (88%), and both marked 15 cases implausible. False positives inherit the length confound, and false negatives occur when a wrong answer reuses context entities, which keeps overlap high. Two audited cases (items 2251 and 2436) carry a hallucinated label even though the context directly supports the answer, so they look like benchmark label noise rather than model errors. The audit sample is deliberately error-heavy, so these proportions describe the audit and not the test set.

5.7. Efficiency and the LLM judge

Table 11 breaks down the latency over 200 test samples. The NLI cross-encoder and the spaCy entity pass dominate at 27.1 ms and 21.1 ms median, while the model itself needs 1.3 ms and SHAP 1.9 ms. The total is 61.8 ms at the median, the deployed artifact weighs 0.82 MB, and the marginal cost of a local prediction is about $0.001 per 1,000 calls. The interface reports about one second per interactive request because model overhead is not amortized as in the batch run.

Table 11
Per-component latency in ms (n = 200). Components are timed in separate instrumented passes over the same samples, so the component means need not sum to the end-to-end median. Retrieval and judge calls are excluded.

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Total per sample	61.77		

Table 12 compares the deployed model with GPT 5.6 Luna on the same 200 samples. The judge reaches an F1 of 0.841 against 0.985 for the model, and the two agree on 84.5% of the cases. The paired McNemar test is highly significant ($p = 1.6 \times 10^{-5}$), with 28 cases where the judge is wrong and the model is right against 3 where the roles are reversed. The judge costs $0.105 per 1,000 predictions and needs 1.3 seconds at the median, while the model costs about $0.001 per 1,000 and runs in 61.8 ms.

Table 12
LLM-as-judge vs. EC-XGB on 200 test samples.

Model	Acc	P	R	F1	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.860	0.974	0.740	0.841	1310	$0.105
EC-XGB (proposed)	0.985	0.980	0.990	0.985	62	≈$0.001

5.8. Error analysis

The 3,000-sample test set contains 48 errors, 12 false positives and 36 false negatives, an error rate of 1.6%. A heuristic review of 20 sampled errors (10 false positives and 10 false negatives) assigns them to two categories. Short-answer ambiguity covers very short answers where neither the lexical nor the NLI evidence is decisive, and label ambiguity covers answers that are semantically plausible under both labels. Table 13 reports the counts. The sample is balanced across the two error types, so the counts describe the reviewed cases rather than the full error population. Both categories trace back to the construction of the benchmark, where short answers are frequently hallucinated by design.

Table 13
Error taxonomy for 20 sampled misclassifications.

Category	FP	FN
Label ambiguity	7	4
Short-answer ambiguity	3	6

The length distribution behind those errors is the more informative signal. Table 14 bins the full test set by answer length. The hallucinated share is 2.1% among one-word answers and rises to 99.2% above 17 words, so answer length is close to a perfect proxy for the label in HaluEval. The model reproduces the pattern. The mean score climbs from 0.024 to 0.993 across the bins. Two-thirds of all false positives fall in the 5-8 word bin at a 10.6% false-positive rate, and false negatives concentrate in the 1-4 word bins at about an 18% rate. This mechanism explains the zero-shot transfer result. Natural RAGTruth responses are full sentences, so the model reads their length as evidence of hallucination, which is why recall reaches 1.0 and AUROC stays near 0.5 on that corpus. The deployed interface responds to this failure mode by leading with per-claim evidence verdicts and presenting the model score as a secondary signal. Two shortcut controls frame the headline result. A logistic regression on the two length features alone reaches F1 = 0.9457 and AUROC = 0.9759, so a large part of the benchmark signal is answer length. On a length-balanced subset with 247 rows from each of the six length bins, EC-XGB reaches F1 = 0.9867 and AUROC = 0.9987. The model therefore holds when the length mix is neutralized. The overlap heuristic in Table 4 plays the same role for lexical overlap (F1 = 0.9352), and the full model's margin over both shortcuts is reported as such.

Table 14
Test-set errors by answer length, mean over seeds (17+ contains only two negative rows).

Words	Rows	Hallucinated	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3–4	474	0.127	0.002	0.178	0.126
5–8	641	0.902	0.106	0.031	0.888
9–16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

5.9. Discussion and limitations

Around 85% of HaluEval is generated with a sampling-then-filtering procedure [20], so its hallucination patterns are synthetic by construction. The zero-shot transfer result on RAGTruth shows how far in-distribution performance is from natural responses. Length and overlap statistics are highly informative on HaluEval but may not carry over, and the binary label leaves graded severity out of scope. The audit found two cases where the hallucinated label contradicts the context, so some label noise remains. Every feature in the pipeline is English-centric, and the supervised model now trains on HaluEval plus the RAGTruth non-QA tasks, yet domain adaptation remains open work at this scale. Stronger neural baselines, such as a fine-tuned encoder detector, were not evaluated, and the LLM judge is the strongest external comparison in this study. Under domain shift EC-XGB trades recall for precision on unseen corpora. On FaithBench,

the flag rate falls from 99.5% to 7.3% while the F1 drops because the model becomes conservative. The RAGTruth all-task gains also include task types that the multi-source training saw, so they measure adaptation rather than pure zero-shot transfer, and the held-out QA task improves mainly in calibration. On the source benchmark the raw probabilities are already well calibrated, and the display calibrator cuts the target ECE from 0.726 to 0.126. The calibrated score is also style-sensitive in conversational use, so the interface leads with the evidence verdicts and presents the model score as a secondary signal. Four established directions fit the verification layer next. RAGAs scores claim support and answer relevance without human review [34], and RAGChecker diagnoses retrieval and generation at the claim level [35]. VeriScore scores verifiable claims in long-form answers [36], and small verifier models make claim-level checks cheap enough for every answer [37].

A hallucination-risk score predicts risk, not truth, and it must not be treated as a fact-checker. Over-reliance on automated detectors creates false trust, especially when scores are confidently wrong out of domain, which is exactly what the RAGTruth result demonstrates. Detection quality varies across domains and languages, and the current system is English-only. The interface therefore labels scores as risk estimates and shows a training-data warning with every prediction.

6. Conclusion

Hallucination detection needs to be cheap, fast, and explainable before it is useful in practice. This study presented EC-XGB, an evidence-consistent XGBoost risk predictor. It combines 26 engineered evidence-consistency features with eight claim-level NLI aggregates, log-scaled length features under monotone constraints, multi-source training, calibrated probabilities, and SHAP explanations. On HaluEval QA it reaches an F1 of 0.9855 and an AUROC of 0.9984, on par with the standard tuned XGBoost on a saturated benchmark. It makes fewer false positives at the same operating point. The gain appears under domain shift. The standard model flags 99.9% of natural RAGTruth answers as risky and EC-XGB flags 61.3%, with the AUROC rising from 0.475 to 0.582 and the ECE falling from 0.635 to 0.278. The deployed display calibrator cuts the RAGTruth QA ECE from 0.726 to 0.126 and needs a labeled target sample (5,034 rows here). The displayed score is a risk estimate, not a fact-checking verdict. One analysis takes 61.8 ms at the median and costs about $0.001 per 1,000 predictions. That is roughly 100 times less than the GPT 5.6 Luna judge, which reaches a lower F1 on the same 200 samples.

The negative results are reported with the same care as the positive ones. The raw model needs no recalibration on HaluEval, and the entity and hedging features add no measurable value. The held-out RAGTruth QA task stays near an F1 of 0.30 even after multi-source training, while the conservative FaithBench regime trades F1 for calibration. The paper therefore frames the supervised risk model as a benchmark and adaptation estimator, and the deployed system adds claim-level verification with evidence retrieval. Future work targets multicalibration under stronger shift, conformal prediction [38], per-domain threshold transfer, shortcut-controlled evaluation on unseen tasks, and neural baselines for comparison.

Declarations

- **Funding:** The authors received no specific funding for this work.
- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Data availability:** The raw evaluation corpora are not redistributed. The repository ships the download scripts, source revisions, SHA-256 hashes, and license notes for HaluEval (MIT), RAGTruth (MIT), and FaithBench (CC BY-NC-SA 4.0). Every split and feature matrix can be regenerated from the original sources. Model artifacts, split indices, and the frozen experiment manifest are included with the code release.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available in the project repository.
- **Ethical approval:** The study uses publicly released benchmark datasets under their original licenses. It involves no human participants and no personal data, so ethical approval is not required.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Saiful Alam Sabbir:** Writing - review & editing. **MD. Miraz Ahamed:** Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

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